

Software Engineering and Project Management Lab Experiment No: - 09

Aim: To Study and Implement a Container using Docker.

Aim: To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers.

Theory:

A container is a standard unit of software that packages up code and all its dependencies so the application runs quickly and reliably across different computing environments. Docker provides a platform for developing, shipping, and running containers easily.

Key Features of Containers:

- ❖ Isolation: Each container runs in its own isolated environment.
- ❖ Portability: Containers can run consistently across any system.
- ❖ Lightweight: Share OS kernel, making them more efficient than virtual machines.
- ❖ Scalable: Easily deploy and replicate containers across clusters.
- ❖ Fast Deployment: Start in seconds, ideal for CI/CD and agile development.
- ❖ Reproducible: Same environment from development to production.

Docker as a Container Platform

Docker enables you to create and manage containers using simple commands. It allows you to containerize applications and services with ease and deploy them in any environment.

Key Docker Container Components:

Component	Purpose
Dockerfile	Blueprint to build a Docker image
Docker Image	Snapshot of the app and its dependencies
Docker Container	Runnable instance of a Docker image
Docker CLI	Command-line tool to interact with Docker
Docker Daemon	Background process managing Docker containers
Docker Hub	Online repository to store and share images

Demonstration of Running a Container using Docker (Theoretical Steps)

1. Install Docker
 - i. Download and install Docker Desktop from <https://www.docker.com>.
 - ii. Start Docker on your local machine.
 - iii. Verify installation by running:

docker --version

2. Pull a Base Image
 - i. Use Docker Hub to pull a popular image (e.g., Ubuntu, Nginx, Python):

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docker pull ubuntu

3. Run a Container from an Image

- i. Use the docker run command:

docker run -it ubuntu

- ii. This opens an interactive terminal session in the container.
- iii. Run Linux commands inside the container (e.g., ls, pwd, apt update).

4. Exit the Container

- i. Type exit to close the session and stop the container.

5. List Running and Stopped Containers

- i. View active containers:

docker ps

- ii. View all containers (including stopped ones):

docker ps -a

6. Remove Containers

- i. Stop the container (if still running):

docker stop <container_id>

- ii. Remove the container:

docker rm <container_id>

7. Run a Web Server Container (Optional)

- i. Run a web server (e.g., Nginx):

docker run -d -p 8080:80 nginx

- ii. Open <http://localhost:8080> in browser to see the Nginx welcome page.

Use Case Example:

- ❖ Running a Linux environment on any OS for testing
- ❖ Hosting a web server like Apache or Nginx inside a container
- ❖ Developing and testing Python, Node.js, or Java apps
- ❖ Containerizing databases like MySQL or MongoDB for quick use
- ❖ Experimenting with new tools and languages without affecting host OS
- ❖ Education and training in DevOps or system admin practices
- ❖ Deploying a standalone app for development or demonstration

Implementation:

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```
ubuntu@ip-172-31-40-218: ~$
ubuntu@ip-172-31-40-218:~$ systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2023-04-10 19:46:05 UTC; 18min ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 684 (dockerd)
      Tasks: 12
     Memory: 141.2M
        CPU: 3.736s
    CGroup: /system.slice/docker.service
            └─684 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Apr 10 19:46:01 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:01.899794323Z" level=info msg="ccResolverWrapper: se
Apr 10 19:46:01 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:01.899956895Z" level=info msg="ClientConn switching >
Apr 10 19:46:02 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:02.339992511Z" level=info msg="[graphdriver] using p
Apr 10 19:46:03 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:03.665306795Z" level=info msg="Loading containers: s
Apr 10 19:46:04 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:04.873139021Z" level=info msg="Default bridge (docke
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.081644328Z" level=info msg="Loading containers: d
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.543882435Z" level=info msg="Docker daemon" commit
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.547797680Z" level=info msg="Daemon has completed
Apr 10 19:46:05 ip-172-31-40-218 systemd[1]: Started Docker Application Container Engine.
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.743749833Z" level=info msg="API listen on /run/do
lines 1-22/22 (END)

ubuntu@ip-172-31-40-218:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
ubuntu@ip-172-31-40-218:~$ docker images
REPOSITORY    TAG       IMAGE ID   CREATED   SIZE
ubuntu@ip-172-31-40-218:~$

ubuntu@ip-172-31-40-218:~$
ubuntu@ip-172-31-40-218:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-40-218:~$ mkdir my-website
ubuntu@ip-172-31-40-218:~$ cd my-website/
ubuntu@ip-172-31-40-218:~/my-website$ wget https://www.free-css.com/assets/files/free-css-templates/download/page290/wave-cafe.zip
--2023-04-10 20:06:14-- https://www.free-css.com/assets/files/free-css-templates/download/page290/wave-cafe.zip
Resolving www.free-css.com (www.free-css.com)... 217.160.0.242, 2001:8d8:100f:f000::28f
Connecting to www.free-css.com (www.free-css.com)|217.160.0.242|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11896390 (11M) [application/zip]
Saving to: 'wave-cafe.zip'

wave-cafe.zip      100%[=====>] 11.34M  6.08MB/s   in 1.9s

2023-04-10 20:06:17 (6.08 MB/s) - 'wave-cafe.zip' saved [11896390/11896390]

ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ unzip wave-cafe.zip

inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.ttf
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.woff
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.woff2
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.eot
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.svg
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.ttf
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.woff
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.woff2
creating: 2121_wave_cafe/img/
inflating: 2121_wave_cafe/img/about-1.png
inflating: 2121_wave_cafe/img/about-2.png
inflating: 2121_wave_cafe/img/hot-americano.png
inflating: 2121_wave_cafe/img/hot-cappuccino.png
inflating: 2121_wave_cafe/img/hot-espresso.png
inflating: 2121_wave_cafe/img/hot-latte.png
inflating: 2121_wave_cafe/img/iced-americano.png
inflating: 2121_wave_cafe/img/iced-cappuccino.png
inflating: 2121_wave_cafe/img/iced-espresso.png
inflating: 2121_wave_cafe/img/iced-latte.png
inflating: 2121_wave_cafe/img/smoothie-1.png
inflating: 2121_wave_cafe/img/smoothie-2.png
inflating: 2121_wave_cafe/img/smoothie-3.png
inflating: 2121_wave_cafe/img/smoothie-4.png
inflating: 2121_wave_cafe/img/special-01.jpg
inflating: 2121_wave_cafe/img/special-02.jpg
inflating: 2121_wave_cafe/img/special-03.jpg
inflating: 2121_wave_cafe/img/special-04.jpg
inflating: 2121_wave_cafe/img/special-05.jpg
inflating: 2121_wave_cafe/img/special-06.jpg
inflating: 2121_wave_cafe/index.html
creating: 2121_wave_cafe/js/
```

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```
ubuntu@ip-172-31-40-218: ~/my-website
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe  wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ cd 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ ls
css  fontawesome  img  index.html  js  video
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cp -R * ../
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cd ..
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe  css  fontawesome  img  index.html  js  video  wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ rm -rf wave-cafe.zip 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
css  fontawesome  img  index.html  js  video
ubuntu@ip-172-31-40-218:~/my-website$ nano Dockerfile
```

```
GNU nano 6.2 Dockerfile
FROM httpd:2.4
COPY . /usr/local/apache2/htdocs/
```


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```
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ cd 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ ls
css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cp -R * ../
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cd ..
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe css fontawesome img index.html js video wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ rm -rf wave-cafe.zip 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website$ nano Dockerfile
ubuntu@ip-172-31-40-218:~/my-website$ ls
Dockerfile css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website$ docker build . -t my-website:latest
Sending build context to Docker daemon 13.61MB
Step 1/2 : FROM httpd:2.4
2.4: Pulling from library/httpd
f1f26f570256: Pull complete
a6b093ae1967: Pull complete
6b400bbb27df: Pull complete
6e310dd059b6: Pull complete
471cb5914961: Pull complete
Digest: sha256:4055b18d92fd006f74d4a2aac172a371dc9a750eaa78000756dee55a9beb4625
Status: Downloaded newer image for httpd:2.4
--> dcl95e13784
Step 2/2 : COPY . /usr/local/apache2/htdocs/
--> 7d48427f5e2f
Successfully built 7d48427f5e2f
Successfully tagged my-website:latest
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ clear
```

```
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
my-website latest 7d48427f5e2f 15 seconds ago 159MB
httpd 2.4 dcl95e13784 4 days ago 145MB
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker run -d -p 80:80 my-website:latest
e0a6d7f3ab6718alb648d9b5f00dccc89e846d1fe12bd568ce9b1412fc0d3c9da
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
e0a6d7f3ab67 my-website:latest "httpd-foreground" 8 seconds ago Up 7 seconds 0.0.0.0:80->80/tcp, :::80->80/tcp
trusting_rosalind
ubuntu@ip-172-31-40-218:~/my-website$
```

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Conclusion: We have successfully understood Docker Architecture and Container Life Cycle, installed Docker and executed docker commands to manage images and interact with containers.

LO Mapping: *LO is mapped*